

# **Buried Structure Detection using Ground Penetrating Radar: Case Study Bunker at ITB Campus**

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## **Abstract**

Liquefaction occurs when vibrations or water pressure within a mass of soil cause the soil particles to lose contact with one another. Being located at the tectonic activity area, Indonesia has to cope with constant risk of earthquake. High frequency of earthquake leads into another natural disaster such as liquefaction. In Indonesia, liquefaction happened in September as the result of the Palu 7.4 Mw Earthquake. This liquefaction caused 1700 buildings and houses swallowed up. In this study, we use Ground Penetrating Radar (GPR) to detect the bunker as an analogy for the detection of buried structure caused by liquefaction. Ground Penetrating Radar (GPR) is one of the geophysical methods that can use to study shallow subsurface of the earth. A detailed GPR survey was conducted in the Metrology Laboratory of Bandung Institute of Technology. The aim of this research is too determine the depth of the bunker and compare the anomalous response of the measurement with the results of forward modeling The target of the GPR survey where the bunker is buried at the depth of ~1 meters. We obtained the amplitude contrast at around 0.5-1.5 meters depth which is consistent with the depth of buried structure. The 250 MHz GPR showed the diffraction pattern in the processed radargram, especially at the edge of the structure. In addition, the result of forward modeling of homogenous subsurface and bunker (concrete and air) in subsurface match with the processed field data.

*Keywords: Liquefaction, buried structure, GPR, earthquake, subsurface, bunker*

## INTRODUCTION

Indonesia is one of the countries in the world that has a vast area and a very diverse cultural diversity. Indonesia as a country that has a history of the kingdom in ancient times, so many relics of objects or historic buildings buried under the surface and have not detected its existence. In addition, Indonesia is also a country that has high cases of natural disasters. One of the most frequently disaster happen in Indonesia is Earthquake. According to BMKG, Indonesia has at least 4500 earthquake in a year. Recently, an earthquake with 7.4 Mw occurred at Palu City on 28 September 2018. This earthquake causing a very rare phenomenon called liquefaction. Soil liquefaction occurs when a saturated or partially saturated soil substantially loses strength and stiffness in response to an applied stress such as shaking during an earthquake or other sudden change in stress condition, in which material that is ordinarily a solid behaves like a liquid. (Allen Hazen, 1918). The sand layer turns into a liquid, so it cannot support the building load inside or above it, which is caused by a cyclic load at the time of the earthquake so that the pore water pressure increases near or exceeds the vertical stress. As the water pressure increases, the distance between the sand particles becomes increasingly tenuous, so that the total strength decreases dramatically. This resulted in the buildings above it being able to enter the soil, making it difficult to find the buildings and the disaster victims.

Ground Penetrating Radar or GPR is one of the geophysical methods to identify subsurface structures with relatively shallow depths. The principle of the GPR method is basically to utilize electromagnetic wave propagation which is a response of contrast permittivity between the target and its environmental material (Baker et al., 2007). GPR can also be used to determine the conditions and characteristics of underground surfaces without drilling or digging soil. As with most radar systems, the GPR system consists of senders (transmitters), namely the antenna connected to the pulse source, and the receiver (receiver), namely the antenna connected to the signal processing unit and the image. To determine the type of the antenna, the transmitted signal and signal processing method depends on several things, such as the type of object to be detected, depth of object and the characteristics of soil medium.

The GPR method also has been used to detect many underground objects, including Conyer's research about structure detection at Ceren Site, El Salvador in 1995. Theoretically, the GPR method will work well if the object has a contrast in dielectric constant. The buried structure commonly built by a concrete. The concrete has different dielectric constant compared to the soil. Therefore, we can use the GPR method to search for underground objects including buried historic building, buried structure caused by liquefaction, or anything buried underground. The aim of this research is to determine the depth of the bunker and compare the anomalous response of the measurement with the results of forward modeling

## DATA AND METHOD

### Forward Modeling

Before taking the data, we do the forward modeling first. The forward modeling including the homogenous model, the 2 layers model, the 3 layers model, and the model that represent the bunker. The model parameters we used in the forward modeling are resistivity of the medium, dielectric constant, and the magnetic permeability. The medium we used in the forward modeling are soil (resistivity = 1000 Ohm. m, dielectric constant = 5, magnetic permeability = 1 H. m<sup>-1</sup>), concrete (resistivity = 5000 Ohm. m, dielectric constant = 4.5, magnetic permeability = 10<sup>-3</sup> H. m<sup>-1</sup>), and air (resistivity = 10<sup>15</sup> Ohm. m, dielectric constant = 1, magnetic permeability = 10<sup>-6</sup> H. m<sup>-1</sup>)

### Data Acquisition

The GPR observation was conducted at the field above the Industrial Metrology Laboratory of the Faculty of Mechanical Engineering and Aerospace of the Bandung Institute of Technology (FTMD - ITB) in October 2018. The settings of the observation field is presented in Figure 1.



(a)



(b)



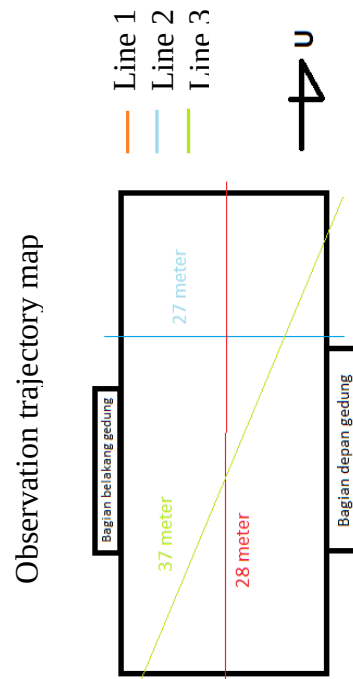
(c)



(d)

**FIGURE 1.** The settings of the observation field. (a) View of the entrance of Industrial Metrology Laboratory (east-side of the field); (b) View of the observation field from Jalan II ITB (south-side of the field); (c) View of the observation field from Jalan D ITB (west-side of the field); (d) View of the observation field from the Telecommunication Department Building (north-side of the field).

The observation used MALA shielded antenna with the frequency of 250 MHz. The assumption of the top of the bunker is located at the depth of roughly 0.8 meters up to 1.5 meters. We used three observation lines in hope to find the bunker structure. Line 1 has a length of 27 meters with orientation of north-south, line 2 has a length of 37 meters with orientation of northeast-southwest, and line 3 has a length of 28 meters with orientation of east-west. The observation trajectory map is presented in Figure 2.



**FIGURE 2.** Observation trajectory map

## Data Processing

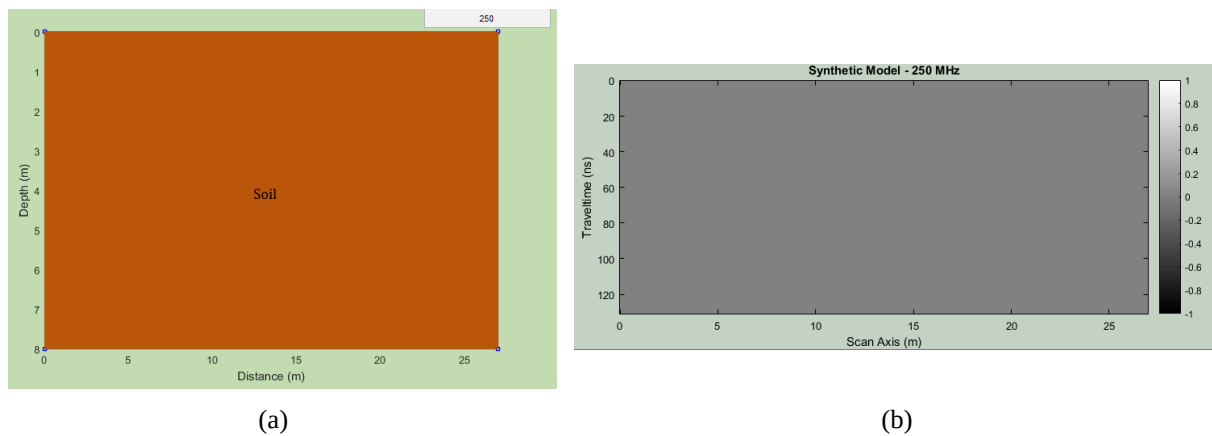
Data processing in this study is divided into two. The first part is forward modeling using different models, ranging from the simple models to the nearly-identical model of the bunker. Forward modeling is used in the survey design to foresee the expected result and later may be used as quality control of the acquired data from observation. The second part is data processing using *ReflexW* software. In this part, the data processing consisted of four stages. The first stage is dewow filter, used to eliminate noise with low frequency. The second stage is butterworth bandpass filter, used to improve signal-to-noise ratio by filtering signals recorded by GPR antennas based on the frequency range used, which is between 100 Hz to 300 Hz for this study. The third stage is automatic gain control, to improve the strength of the electromagnetic amplitudes to improve data quality. And the last stage is time cut, to focus the results of the processing data at the desired target depth, around 10 meters for this study.

## RESULT AND DISCUSSION

### Forward Modeling

Forward modeling is done before data acquisition and is included in the survey design stage to see what the desired results are. In addition, forward modeling is the quality control for data obtained from the acquisition. In this research, forward modeling was carried out with several models from simple to near real conditions. This is done to make it easier to control the quality of the data produced.

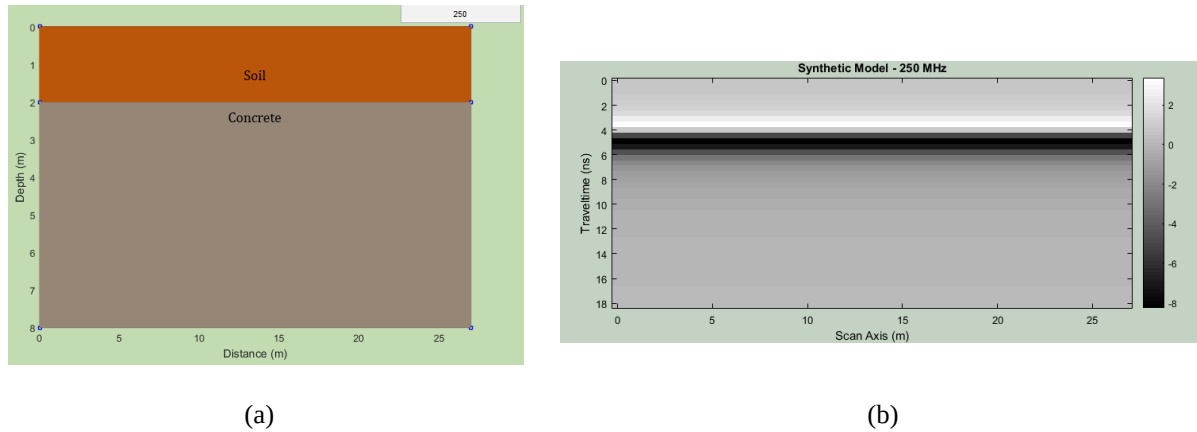
The first model is a model with one layer that is homogeneous. The results of forward modeling for this model along with the response are shown in Figure 3. Visible radargram produced has a homogeneous value.



**FIGURE 3.** (a) Synthetic Model of homogeneous background, (b) model response

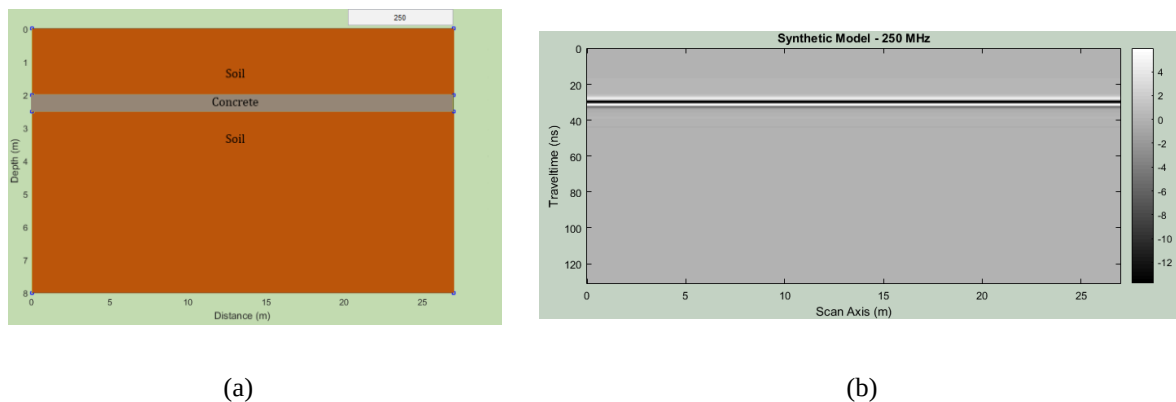
The second model is a two-layer model of soil and concrete. The soil layer is on the surface to a depth of 2 meters, while the concrete layer is at a depth of 2 meters to 8 meters. The results of the forward modeling for the

model are shown in Figure 4. The generated radar shows that there is a change in the value at the very large boundary.



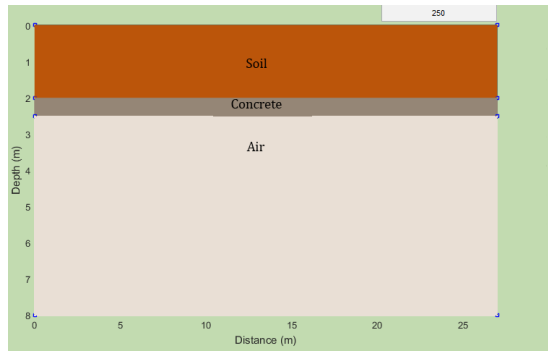
**FIGURE 4.** (a) Synthetic model of background and concrete, (b) model respond

The third model is a three-layer model, namely soil, concrete and soil. The model in the form of 8 meters thick soil inserted by a concrete column is at a depth of 2 meters with a height of 0.5 meters. The future modeling results for this model are shown in Figure 5. The resulting radargram shows the return value at the boundary of the coating.

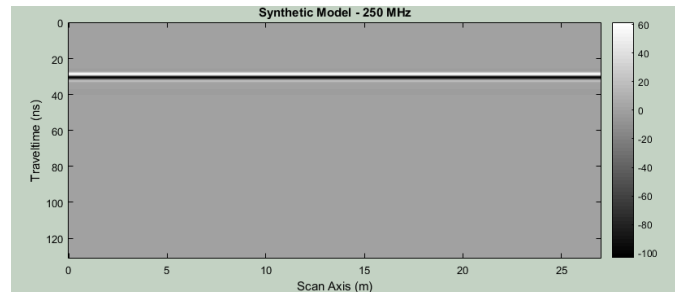


**FIGURE 5.** (a) Synthetic model of soil and concrete layer, (b) model respond

The fourth model is a three-layer model, namely soil, concrete and air. This model is a 2 meter thick layer of soil, a layer of concrete with a thickness of 0.5 meters at the bottom, then there is air below it to a depth of 8 meters. The results of forward modeling for this model are shown in Figure 6 and Figure 7. Note that the Figure 7 represent the real model of the bunker. Visible radargrams that are produced show a change in value at the boundary of the coating.

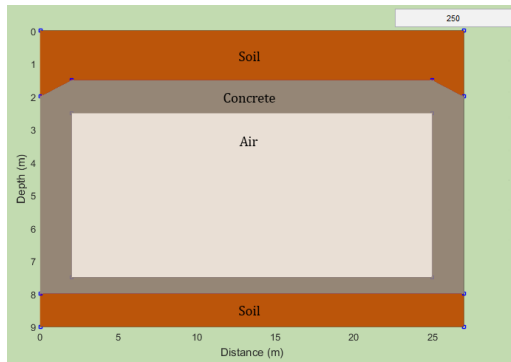


(a)

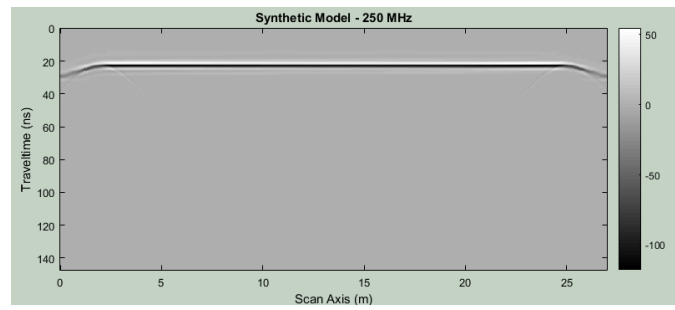


(b)

**FIGURE 6.** (a) Synthetic model of soil, concrete, and air, (b) model respond



(a)

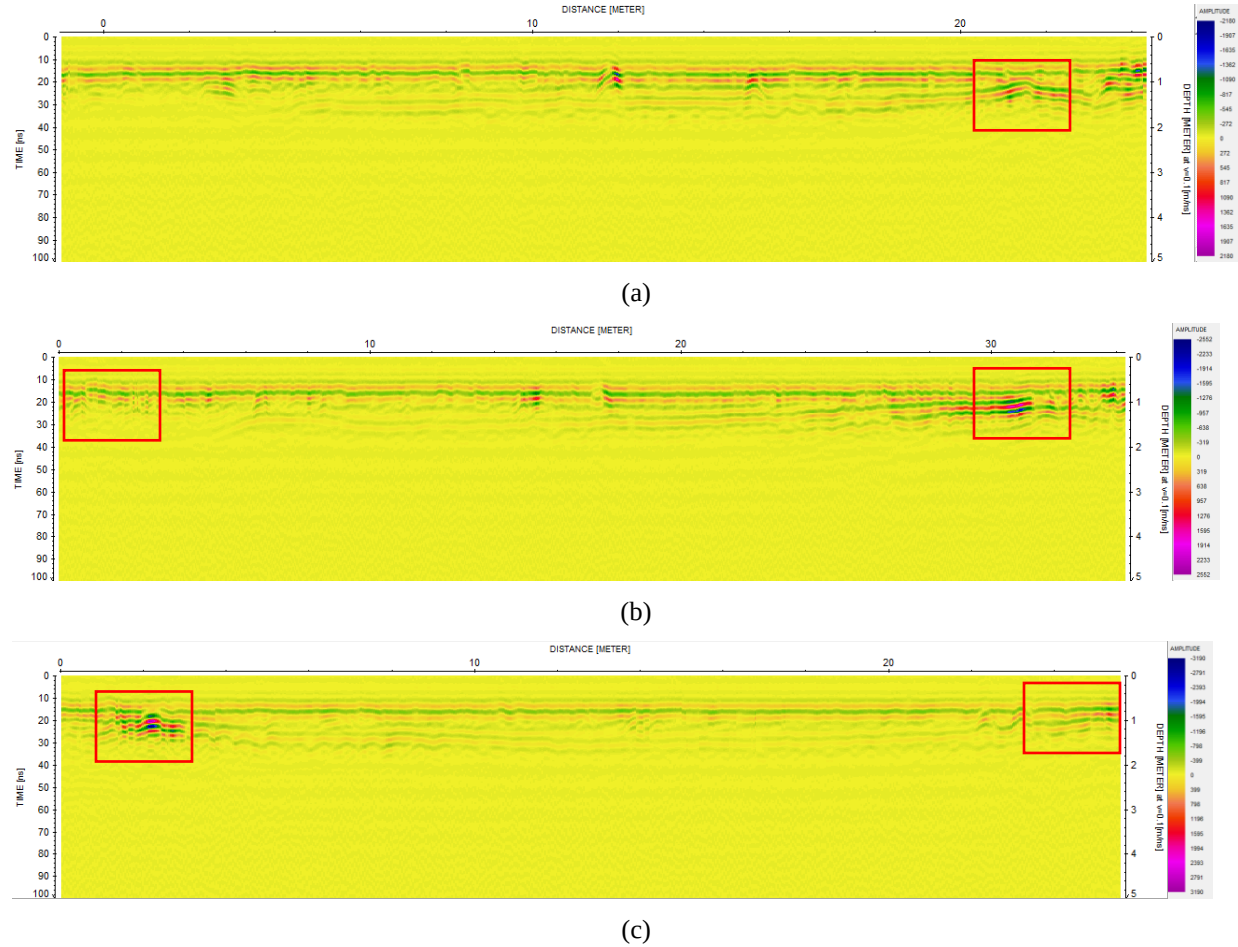


(b)

**FIGURE 7.** (a) Synthetic model of the real bunker, (b) model respond

## Data Processing Results with ReflexW

After processing the data, 3 radargrams are obtained from each measurement line (Figure 2). The radargrams show in general, there is a clear reflector.



**FIGURE. 6** (a) The processed radargram for line 1 (N-S) (b) The processed radargram for line 2 (SW-NE); (c) The processed radargram for line 3 (W-E). The red blocks show the edge of the structure.

Geologically, the measurement area is a building buried by a 0.8-1.5 meter thick layer of soil. In some places air ducts were found on the surface. From the radargram data, it can be seen that there is a reflector at a depth of about 0.6-1.3 meters. This can be interpreted as the top layer of concrete in this building. The thickness of the concrete layer can be clearly seen from radargram data which is around 0.5-0.7 meters. In line 1, there is a strong reflector in the south while in the line 2 is in the northeast. This can be interpreted as building construction located on the front of the building (Figure 2). Whereas in line 3 there is a strong reflector on the western part that can be interpreted as building construction located behind the building.

## CONCLUSION

Top layer of the building can detected from the radargram data, there is a reflector at a depth of about 0.6-1.3 meters with thickness of the concrete can be clearly seen which is around 0.5-0.7 meters. Another building construction where is located in front of building (Figure 2) also can be interpreted by strong reflector from line 1 (south) and line 2 (northeast). The result also match with the forward modeling of the bunker synthetic model. So, it is possible to detect a buried structure due to liquefaction or buried historic building or even anything buried underground near the surface using GPR method.

## REFERENCES

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